

The maximal positive coefficient of a Hermite polynomial

Exact maximization, asymptotic growth, and lattice fluctuations

A research note on OEIS A277280

20 September 2026

Abstract

Let $a(n)$ be the largest coefficient of the physicists' Hermite polynomial $H_n(x)$, with signs taken into account. We prove the full asymptotic equivalent

$$a(n) \sim \frac{\exp(\sqrt{2n} - \frac{1}{2})}{\sqrt{\pi} (2n)^{1/4}} \left(\frac{2n}{e} \right)^{n/2}.$$

The proof includes an exact, integer-arithmetic formula for the uniquely maximizing exponent. A factorization reduces the apparent quartic comparison between adjacent positive coefficients to a quadratic inequality. Uniform Stirling expansions then give three explicit correction polynomials. The first correction depends on a bounded rounding displacement and genuinely oscillates; its sharp lower and upper limits and its limiting distribution are determined. We also prove $a(n+1)/(a(n)\sqrt{n}) \rightarrow \sqrt{2}$, describe its first-order fluctuations, and compare the answer with the maximum absolute coefficient. Exact Python implementations, independent finite checks, high-precision numerical tables, and reproducible figures accompany the note.

Scope. The sequence definition and normalization are taken from OEIS and NIST DLMF. The exact maximization, asymptotic expansions, and fluctuation results below are proved in the text rather than inferred from numerical fitting. No claim of a complete literature search or historical priority is made.

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1 The sequence and the normalization

OEIS A277280 is the maximal coefficient of $H_n(x)$, where coefficients retain their signs [1]. The entry was introduced by Vladimir Reshetnikov in October 2016. The relevant Hermite polynomials are the *physicists'* polynomials, not the differently normalized probabilists' polynomials. Their explicit expansion is

$$H_n(x) = n! \sum_{j=0}^{\lfloor n/2 \rfloor} \frac{(-1)^j 2^{n-2j} x^{n-2j}}{j! (n-2j)!}, \quad (1)$$

as in DLMF 18.5.13 [2]. In particular,

$$H_5(x) = 32x^5 - 160x^3 + 120x, \quad a(5) = 120,$$

not 160. Zero coefficients do not affect the maximum, since the leading coefficient 2^n is positive.

A coefficient in (1) is positive precisely when j is even. Writing $j = 2k$ and $d = n - 4k$ gives

$$a(n) = \max_{0 \leq k \leq \lfloor n/4 \rfloor} \frac{n! 2^{n-4k}}{(2k)! (n-4k)!} = \max_{d \in \mathcal{D}_n} C_n(d), \quad (2)$$

where

$$\mathcal{D}_n = \{d \in \mathbb{Z} : 0 \leq d \leq n, d \equiv n \pmod{4}\}, \quad C_n(d) = \frac{n! 2^d}{((n-d)/2)! d!}. \quad (3)$$

The distinction between a lattice of spacing 4 and one of spacing 2 is the entire sign constraint. It will affect lower-order terms, but not the leading asymptotic constant.

The first terms, indexed from $n = 0$, are

$$1, 2, 4, 8, 16, 120, 720, 3360, 13440, 48384, 302400, 2217600, \dots$$

The closely related sequence A277281 takes the maximum *absolute* coefficient instead [4]; Section 8 compares the two.

2 An exact solution of the discrete maximization

The maximizing coefficient can be found without constructing a polynomial, scanning its coefficients, or evaluating any floating-point square root.

Theorem 2.1 (Exact maximizing exponent). *For every integer $n \geq 0$, set*

$$k_n = \left\lfloor \frac{2n + 6 - \lfloor \sqrt{8n + 21} \rfloor}{8} \right\rfloor, \quad d_n = n - 4k_n. \quad (4)$$

Then d_n is the unique maximizing exponent in (2), and

$$a(n) = \frac{n! 2^{d_n}}{(2k_n)! d_n!}. \quad (5)$$

Equivalently, if

$$r_n = \frac{\sqrt{8n + 21} - 7}{2}, \quad (6)$$

then d_n is the smallest allowed exponent strictly greater than r_n . In fact, $r_n < d_n < r_n + 4$.

Proof. For two consecutive allowed exponents $d, d+4$, put $m = (n-d)/2$. Cancellation of factorials gives

$$\frac{C_n(d+4)}{C_n(d)} = \frac{16m(m-1)}{(d+1)(d+2)(d+3)(d+4)} = \frac{4(n-d)(n-d-2)}{(d+1)(d+2)(d+3)(d+4)}. \quad (7)$$

The difference between the denominator and numerator has the useful factorization

$$\begin{aligned} & (d+1)(d+2)(d+3)(d+4) - 4(n-d)(n-d-2) \\ &= (d^2 + 7d + 7 - 2n)(d^2 + 3d + 3 + 2n) + 3. \end{aligned} \quad (8)$$

For integer d, n , the first factor on the right is odd, hence nonzero. The second factor is greater than 3 whenever $d, d+4$ are both allowed. Consequently the sign of (8) is exactly the sign of $d^2 + 7d + 7 - 2n$. Thus

$$C_n(d+4) > C_n(d) \iff d^2 + 7d + 7 < 2n. \quad (9)$$

The quadratic on the left is strictly increasing for $d \geq 0$ and crosses $2n$ at r_n . The positive coefficients therefore increase strictly until passing this threshold, and then decrease strictly. This proves uniqueness and identifies the mode.

The endpoint n exceeds r_n . For $n \geq 4$, $r_n > 0$, and for $n = 0, 1, 2, 3$ the asserted mode is checked immediately. Hence the lattice description indeed gives an allowed exponent for every $n \geq 0$. Solving it for k_n gives

$$k_n = \left\lfloor \frac{2n + 7 - \sqrt{8n + 21}}{8} \right\rfloor. \quad (10)$$

Now $8n + 21 \equiv 5 \pmod{8}$, so it is never a square. If $q = \lfloor \sqrt{8n + 21} \rfloor$, then $\sqrt{8n + 21} = q + \varepsilon$ with $0 < \varepsilon < 1$. Replacing the numerator in (10) by $2n + 6 - q$ does not change its quotient's floor: the difference is $1 - \varepsilon \in (0, 1)$ and the latter numerator is an integer. This proves (4). Substitution into (3) proves (5). $\square$

2.1 Localization and a useful rounding interpretation

For $n \geq 1$, write

$$s = s_n = \sqrt{2n}, \quad \delta_n = d_n - s_n + \frac{3}{2}. \quad (11)$$

The exact threshold has the expansion

$$r_n = s_n - \frac{7}{2} + \frac{21}{8s_n} + O(s_n^{-3}). \quad (12)$$

It follows that

$$-2 + \frac{21}{8s_n} + O(s_n^{-3}) < \delta_n < 2 + \frac{21}{8s_n} + O(s_n^{-3}). \quad (13)$$

In particular, $d_n = \sqrt{2n} + O(1)$, while

$$k_n = \frac{n}{4} - \frac{\sqrt{2n}}{4} + O(1).$$

The maximizing term is therefore far below the leading degree n , but not close to a fixed degree either.

Define the exact rounding center

$$h_n = r_n + 2 = \frac{\sqrt{8n+21} - 3}{2}. \quad (14)$$

The exponent d_n is the member of $n + 4\mathbb{Z}$ nearest to h_n . There is no midpoint ambiguity, because r_n is not an integer. This statement is an exact discrete rule, not an assumption that rounding the maximum of a smooth interpolation must always work.

3 The leading asymptotic equivalent

Define, for real $n > 0$,

$$L(n) = \frac{\exp(s_n - \frac{1}{2})}{\sqrt{\pi} s_n^{1/2}} \left(\frac{2n}{e}\right)^{n/2} = \frac{\exp(\sqrt{2n} - \frac{1}{2})}{\sqrt{\pi} (2n)^{1/4}} \left(\frac{2n}{e}\right)^{n/2}. \quad (15)$$

Theorem 3.1 (Full asymptotic growth). *As $n \rightarrow \infty$ through the integers,*

$$\boxed{a(n) = L(n)(1 + O(n^{-1/2}))}. \quad (16)$$

In particular $a(n) \sim L(n)$, with exactly the constant and powers displayed in (15).

Proof. Let $d = d_n = s + O(1)$ and $N = n/2$. Extend factorials by the gamma function. Stirling's expansion [3] first gives

$$\log \frac{\Gamma(n+1)}{\Gamma(N+1)} = \frac{n}{2} \log \frac{2n}{e} + \frac{1}{2} \log 2 + O(n^{-1}). \quad (17)$$

A second application, with the moving displacement $d/2 = O(\sqrt{N})$, gives

$$\log \frac{\Gamma(N+1)}{\Gamma(N-d/2+1)} = \frac{d}{2} \log N - \frac{d^2}{4n} + O(s^{-1}). \quad (18)$$

For completeness, this follows by expanding $\log(N - d/2) = \log N + \log(1 - d/(2N))$ in the two Stirling expressions. The neglected terms have size $O(d/N + d^3/N^2) = O(s^{-1})$; a fixed-shift gamma-ratio formula is not being applied outside its range of uniformity.

Combining (17), (18), and the remaining factor $2^d/\Gamma(d+1)$ yields

$$\begin{aligned} \log C_n(d) &= \frac{n}{2} \log \frac{2n}{e} + \frac{1}{2} \log 2 + d \log s - \log \Gamma(d+1) - \frac{d^2}{4n} + O(s^{-1}) \\ &= \frac{n}{2} \log \frac{2n}{e} + d \log \frac{s}{d} + d - \frac{1}{2} \log(\pi d) - \frac{d^2}{2s^2} + O(s^{-1}). \end{aligned} \quad (19)$$

Because $d = s + O(1)$,

$$d \log \frac{s}{d} + d = s + O(s^{-1}), \quad \log d = \log s + O(s^{-1}), \quad \frac{d^2}{2s^2} = \frac{1}{2} + O(s^{-1}).$$

Therefore

$$\log a(n) = \frac{n}{2} \log \frac{2n}{e} + s - \frac{1}{2} - \frac{1}{2} \log \pi - \frac{1}{2} \log s + O(s^{-1}) = \log L(n) + O(s^{-1}). \quad (20)$$

Exponentiating proves the theorem. $\square$

3.1 Equivalent descriptions of the growth rate

The logarithmic form, with all constant terms made explicit, is

$$\begin{aligned} \log a(n) = & \frac{n}{2} \log n + \frac{n}{2}(\log 2 - 1) + \sqrt{2n} - \frac{1}{4} \log n \\ & - \frac{1}{4} \log 2 - \frac{1}{2} \log \pi - \frac{1}{2} + O(n^{-1/2}). \end{aligned} \quad (21)$$

Thus

$$\frac{\log a(n)}{n \log n} \rightarrow \frac{1}{2}, \quad \frac{a(n)^{1/n}}{\sqrt{n}} \rightarrow \sqrt{\frac{2}{e}}. \quad (22)$$

The factor $\exp(\sqrt{2n})$ is subexponential in n , but it cannot be absorbed into a multiplicative constant. Similarly, the factor $n^{-1/4}$ and the constant $2^{-1/4}/\sqrt{\pi e}$ are needed for a true asymptotic equivalent rather than just logarithmic growth.

3.2 Why the sign restriction does not halve the maximum

The coefficient peak has a width that diverges. To see this directly, consider $C_n(d)$ with the gamma-function extension when necessary and put $d = s + x\sqrt{s}$, where x stays in a fixed bounded interval. The same calculation as (19) is uniform on this scale, and gives

$$\log \frac{C_n(s + x\sqrt{s})}{L(n)} = -\frac{x^2}{2} + O(s^{-1/2}). \quad (23)$$

Indeed, the main varying term is

$$d \log(s/d) + d - s = -\frac{(d-s)^2}{2s} + O\left(\frac{|d-s|^3}{s^2}\right),$$

and the other differences in (19) are $O(s^{-1/2})$. Thus the local profile is Gaussian and its degree scale is $\sqrt{s} = (2n)^{1/4}$. Moving only a bounded distance to a positive-coefficient lattice point changes the height by a relative amount of order $1/s$. Taking half as many candidate degrees does not take half of the peak height.

4 A uniform expansion with explicit rounding corrections

The $O(n^{-1/2})$ term in Theorem 3.1 can be described exactly to first order, but it cannot be replaced by a single constant multiple of $n^{-1/2}$. The rounding displacement δ_n must be retained.

Define

$$P_1(z) = \frac{41}{24} - \frac{z^2}{2}, \quad (24)$$

$$P_2(z) = \frac{z^3}{6} - z^2 + \frac{47z}{24} - \frac{3}{4}, \quad (25)$$

$$P_3(z) = -\frac{z^4}{12} + \frac{z^3}{3} - \frac{35z^2}{24} + \frac{47z}{12} - \frac{8719}{2880}. \quad (26)$$

Theorem 4.1 (Three logarithmic correction terms). *With the exact d_n in Theorem 2.1 and δ_n in (11),*

$$\log \frac{a(n)}{L(n)} = \frac{P_1(\delta_n)}{s_n} + \frac{P_2(\delta_n)}{s_n^2} + \frac{P_3(\delta_n)}{s_n^3} + O(s_n^{-4}). \quad (27)$$

The remainder is uniform over all integers n sufficiently large, including indices at which the maximizing exponent changes.

Proof. Set $d = s + u$, where u is in a fixed bounded interval, and write

$$n = \frac{s^2}{2}, \quad m = \frac{n - d}{2} = \frac{s^2}{4} - \frac{s}{2} - \frac{u}{2}. \quad (28)$$

Use the real-positive Stirling expansion

$$\log \Gamma(x + 1) = \left(x + \frac{1}{2}\right) \log x - x + \frac{1}{2} \log(2\pi) + \frac{1}{12x} - \frac{1}{360x^3} + O(x^{-5}). \quad (29)$$

Its remainder properties are standard [3]. At n, m the arguments are of order s^2 , and at d the argument is of order s . Expand using

$$\begin{aligned} \log n &= 2 \log s - \log 2, \\ \log m &= 2 \log s - \log 4 + \log(1 - 2/s - 2u/s^2), \\ \log d &= \log s + \log(1 + u/s). \end{aligned}$$

Substitution in $\log C_n(d) = \log \Gamma(n + 1) - \log \Gamma(m + 1) - \log \Gamma(d + 1) + d \log 2$ and collection of like powers gives

$$\begin{aligned} \log C_n(s + u) - \log L(n) &= \frac{\frac{7}{12} - \frac{3}{2}u - \frac{1}{2}u^2}{s} \\ &\quad + \frac{\frac{1}{2} + \frac{1}{12}u - \frac{1}{4}u^2 + \frac{1}{6}u^3}{s^2} \\ &\quad + \frac{\frac{97}{360} + \frac{2}{3}u - \frac{13}{12}u^2 - \frac{1}{6}u^3 - \frac{1}{12}u^4}{s^3} + O(s^{-4}). \end{aligned} \quad (30)$$

This is a finite Taylor calculation; the optional symbolic script in the archive verifies the displayed coefficients using exact rational arithmetic.

To justify uniformity, fix $|u| \leq M$. For large s , both logarithm arguments being expanded differ from 1 by less than $1/2$. Their Taylor remainders are then bounded uniformly in u . Expanding $\log m$ through order s^{-5} and $\log d$ through order s^{-4} suffices for the indicated $O(s^{-4})$ remainder after multiplication by $m + 1/2$ and $d + 1/2$. The omitted Stirling remainders are smaller still. Since the exact maximizing $u = d_n - s_n$ is bounded by Section 2, the calculation applies uniformly to it. Finally, $u = \delta_n - 3/2$ in (30) yields (24)–(26). $\square$

Corollary 4.2 (Relative expansion). *The first two relative corrections are*

$$\begin{aligned} \frac{a(n)}{L(n)} &= 1 + \frac{P_1(\delta_n)}{s_n} \\ &\quad + \frac{P_2(\delta_n) + \frac{1}{2}P_1(\delta_n)^2}{s_n^2} + O(s_n^{-3}). \end{aligned} \quad (31)$$

In particular,

$$a(n) = L(n) \left[1 + \frac{\frac{41}{24} - \frac{1}{2}\delta_n^2}{\sqrt{2n}} + O(n^{-1}) \right]. \quad (32)$$

For numerical work it is often preferable to keep the correction in the exponent. Define

$$L_j(n) = L(n) \exp \left(\sum_{i=1}^j \frac{P_i(\delta_n)}{s_n^i} \right), \quad j = 1, 2, 3. \quad (33)$$

Then

$$\frac{a(n)}{L_1(n)} = 1 + O(n^{-1}), \quad \frac{a(n)}{L_2(n)} = 1 + O(n^{-3/2}), \quad \frac{a(n)}{L_3(n)} = 1 + O(n^{-2}).$$

These are asymptotic statements, not assertions that every additional term improves the approximation at every small n .

Remark 4.3 (Extension to any fixed order). Repeating the proof gives polynomials P_j to any prescribed finite order: for every fixed J ,

$$\log \frac{a(n)}{L(n)} = \sum_{j=1}^J \frac{P_j(\delta_n)}{s_n^j} + O(s_n^{-J-1}).$$

More terms of the Stirling series and of the elementary logarithm expansions are sufficient. This is a Poincaré asymptotic expansion; convergence of the infinite series at a fixed n is neither needed nor asserted.

5 Explicit finite-index error bounds

The asymptotic expansion has unspecified but uniform remainder constants. A different approximation supplies simple, explicit bounds at any index where n, m, d are positive. Let

$$S(x) = \left(x + \frac{1}{2} \right) \log x - x + \frac{1}{2} \log(2\pi) + \frac{1}{12x}. \quad (34)$$

For $x > 0$, the alternating-remainder property of Stirling's series gives

$$S(x) - \frac{1}{360x^3} < \log \Gamma(x+1) < S(x) \quad (35)$$

[3]. For the exact maximizing $d = d_n$ and $m = (n - d)/2$, define

$$T_n = S(n) - S(m) - S(d) + d \log 2.$$

Adding the inequalities with the correct subtraction signs proves

$$T_n - \frac{1}{360n^3} < \log a(n) < T_n + \frac{1}{360m^3} + \frac{1}{360d^3}. \quad (36)$$

For A277280 the positivity requirements hold for $n \geq 5$; earlier terms are evaluated exactly. The width of this interval is

$$\frac{1}{360n^3} + \frac{1}{360m^3} + \frac{1}{360d^3} = O(n^{-3/2}).$$

Exponentiation provides corresponding multiplicative bounds. This approach retains the exact factorial arguments rather than expanding them in $1/s$. It is useful when a rigorous finite- n estimate is more important than a short expression in n .

The accompanying function `stirling_log_enclosure` evaluates these analytic endpoints numerically. Its ordinary high-precision floating-point outputs are not themselves directed-rounding interval certificates. A fully certified decimal enclosure would additionally require interval evaluation of the logarithms and of π .

6 The first correction really oscillates

The bound $\delta_n \in [-2 + o(1), 2 + o(1)]$ already restricts the possible corrections. To prove that the endpoints and all intermediate values actually occur as subsequential limits, we establish a little more.

Lemma 6.1 (Elementary square-root equidistribution). *If $\alpha > 0$ and β, γ are fixed real numbers, then the fractional parts of $\sqrt{\alpha m + \beta} + \gamma$ are uniformly distributed on $[0, 1)$ as $m \rightarrow \infty$ through integers for which the expression is defined. For a fixed interval $[u, v) \subset [0, 1)$, the counting error among the first M indices is $O(\sqrt{M})$.*

Proof. For each integer q beyond a fixed initial range, the condition

$$q + u \leq \sqrt{\alpha m + \beta} + \gamma < q + v$$

is equivalent to an interval for m with endpoints

$$\frac{(q + u - \gamma)^2 - \beta}{\alpha}, \quad \frac{(q + v - \gamma)^2 - \beta}{\alpha}.$$

Its length is $\alpha^{-1}(v - u)(2q + u + v - 2\gamma)$, and the number of integers it contains differs from that length by $O(1)$. Summing over $q \leq \sqrt{\alpha M} + O(1)$ gives $(v - u)M + O(\sqrt{M})$. The possibly incomplete last interval contributes only $O(\sqrt{M})$ as well. Dividing by M proves the assertion. $\square$

Theorem 6.2 (Limiting law of the rounding displacement). *The empirical distribution of δ_n tends to $\text{Unif}[-2, 2]$. The same conclusion holds along each fixed residue class of n modulo 4. In particular, the set of subsequential limits of δ_n is exactly $[-2, 2]$.*

Proof. Put $e_n = d_n - h_n$, where h_n is the exact rounding center (14). By nearest-lattice rounding, $e_n \in (-2, 2)$. Fix $r \in \{0, 1, 2, 3\}$ and write $n = 4m + r$. Then

$$\frac{h_n - r}{4} = \sqrt{\frac{m}{2} + \frac{8r + 21}{64}} - \frac{3 + 2r}{8}. \quad (37)$$

Lemma 6.1 applies. If a fractional part is $t \in [0, 1)$, rounding to the nearest integer gives an error of $-t$ on $[0, 1/2)$ and $1 - t$ on $[1/2, 1)$. The image of uniform measure under this map is uniform on $[-1/2, 1/2]$. Multiplying by 4 gives the limiting law for e_n . Finally,

$$\delta_n - e_n = h_n - s_n + \frac{3}{2} = \sqrt{s_n^2 + \frac{21}{4}} - s_n = O(s_n^{-1}) \rightarrow 0.$$

The same limiting law therefore holds for δ_n . The uniform law and (13) also give the asserted complete limit set. $\square$

Corollary 6.3 (Sharp first-order envelopes). *Let $E_n = s_n(a(n)/L(n) - 1)$. Then*

$$\boxed{\liminf_{n \rightarrow \infty} E_n = -\frac{7}{24}, \quad \limsup_{n \rightarrow \infty} E_n = \frac{41}{24}.} \quad (38)$$

Its full set of subsequential limits is $[-7/24, 41/24]$. The same statements hold with E_n replaced by $s_n \log(a(n)/L(n))$.

Proof. Equation (31) gives $E_n = P_1(\delta_n) + O(s_n^{-1})$ uniformly. The image of $[-2, 2]$ under $P_1(z) = 41/24 - z^2/2$ is the stated interval. The logarithmic version follows directly from Theorem 4.1. $\square$

Consequently the $O(n^{-1/2})$ relative error in the leading formula is sharp: it is not $o(n^{-1/2})$. There is no constant c for which

$$a(n) = L(n)(1 + c/s_n + o(s_n^{-1})).$$

An n -dependent rounding correction is genuinely necessary.

More explicitly, the limiting distribution function of E_n is

$$F(x) = \begin{cases} 0, & x < -7/24, \\ 1 - \frac{1}{2}\sqrt{2(41/24 - x)}, & -7/24 \leq x \leq 41/24, \\ 1, & x > 41/24. \end{cases} \quad (39)$$

This is simply the distribution of $41/24 - U^2/2$ for $U \sim \text{Unif}[-2, 2]$. Since the variables involved are uniformly bounded apart from finitely many initial values, it also follows that

$$\frac{1}{N} \sum_{n=1}^N E_n \longrightarrow \frac{41}{24} - \frac{1}{2} \cdot \frac{4}{3} = \frac{25}{24}.$$

The limiting mean is not a substitute for the pointwise oscillatory term.

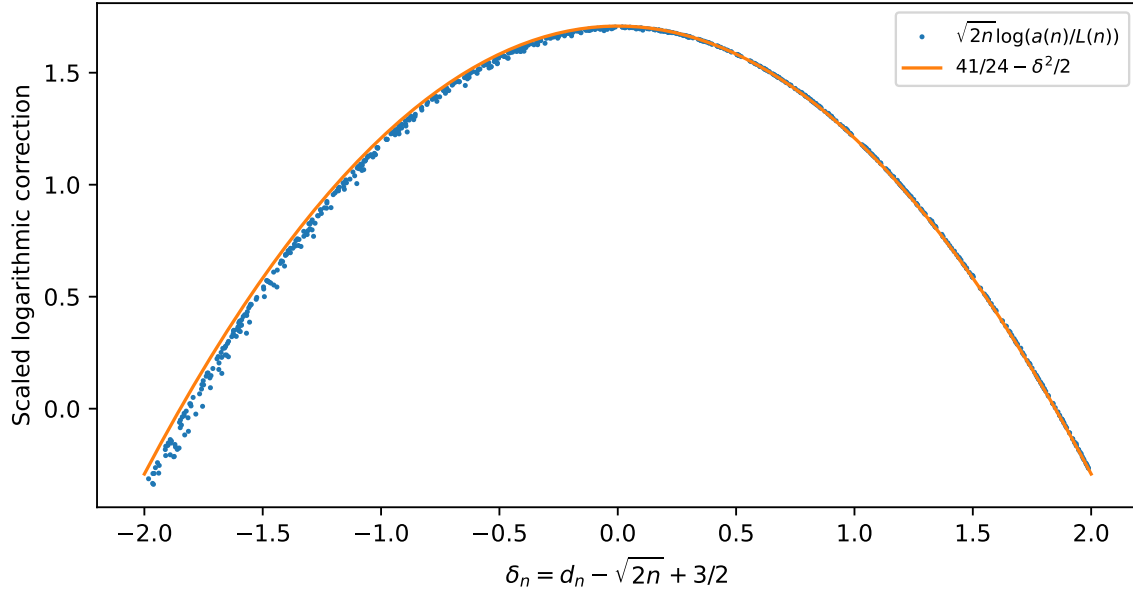


Figure 1: The scaled logarithmic correction versus the exact displacement δ_n , sampled for $1000 \leq n \leq 16000$. The limiting curve is the parabola $P_1(\delta) = 41/24 - \delta^2/2$. All plotted data are newly computed from the formulas in this note.

7 Consecutive ratios and their oscillations

The OEIS entry links to a plot of $a(n+1)/(a(n)\sqrt{n})$ [1]. Its limiting value and its visible fluctuations can both be settled exactly.

Proposition 7.1 (Exact one-step transition). *Write $d = d_n$ and $m = (n - d)/2$. Then*

$$d_{n+1} = \begin{cases} d + 1, & d(d + 1) < 2n + 7, \\ d - 3, & d(d + 1) > 2n + 7. \end{cases} \quad (40)$$

There is no equality case. Correspondingly,

$$\frac{a(n+1)}{a(n)} = \begin{cases} \frac{2(n+1)}{d+1}, & d_{n+1} = d + 1, \\ \frac{(n+1)d(d-1)(d-2)}{8(m+1)(m+2)}, & d_{n+1} = d - 3. \end{cases} \quad (41)$$

Proof. As n increases by 1, the integer $\lfloor \sqrt{8n+21} \rfloor$ increases by either 0 or 1. Indeed the underlying square root increases by less than 1 even at $n = 0$. Thus (4) shows $k_{n+1} - k_n \in \{0, 1\}$, and the only possible exponents are $d + 1$ and $d - 3$.

When both candidates exist, compare them using (9), now at index $n + 1$ and lower degree $d - 3$. The deciding expression is

$$(d - 3)^2 + 7(d - 3) + 7 - 2(n + 1) = d^2 + d - 7 - 2n.$$

Its sign gives (40). If $d < 3$, only the first candidate is valid, and its displayed inequality automatically holds. Equality is impossible because $d(d + 1)$ is even and $2n + 7$ is odd. Finally, cancel factorials in (5) in each of the two cases to obtain (41). $\square$

Theorem 7.2 (Ratio limit and sharp fluctuations). *For $n \geq 1$, let*

$$R_n = \frac{a(n+1)}{a(n)\sqrt{n}}.$$

Then

$$\boxed{R_n \longrightarrow \sqrt{2}.} \quad (42)$$

More precisely, define the continuous, period-4 function whose values on $[-2, 2]$ are

$$H(z) = \begin{cases} \frac{1}{2} - z, & -2 \leq z \leq 1, \\ 3z - \frac{7}{2}, & 1 \leq z \leq 2. \end{cases} \quad (43)$$

Then

$$\sqrt{n}(R_n - \sqrt{2}) = H(\delta_n) + O(n^{-1/2}). \quad (44)$$

In particular,

$$\liminf \sqrt{n}(R_n - \sqrt{2}) = -\frac{1}{2}, \quad \limsup \sqrt{n}(R_n - \sqrt{2}) = \frac{5}{2}. \quad (45)$$

The empirical distribution of the scaled deviations tends to $\text{Unif}[-1/2, 5/2]$.

Proof. Substitute $d = s - 3/2 + \delta_n$, $m = (n - d)/2$, and $s = \sqrt{2n}$ into (41). In the first case, expansion of the rational expression gives

$$R_n = \sqrt{2} \left(1 + \frac{1/2 - \delta_n}{s} + O(s^{-2}) \right).$$

In the second case, the same calculation gives

$$R_n = \sqrt{2} \left(1 + \frac{3\delta_n - 7/2}{s} + O(s^{-2}) \right).$$

The first branch condition is $\delta_n \leq 1 + O(s^{-1})$, and the second is $\delta_n \geq 1 + O(s^{-1})$. Both linear expressions agree at $\delta = 1$. They also agree at the identified endpoints -2 and 2 . Consequently the periodic formula (44) is uniform even near the switching points and at the endpoints in (13).

The limit and the extreme values follow immediately. For the distribution, apply Theorem 6.2. On the first branch, uniform measure of density $1/4$ is mapped with slope of magnitude 1; on the second, with slope 3. Every interior point of $[-1/2, 5/2]$ therefore has total image density $1/4 + 1/12 = 1/3$, which is the uniform density on that interval. $\square$

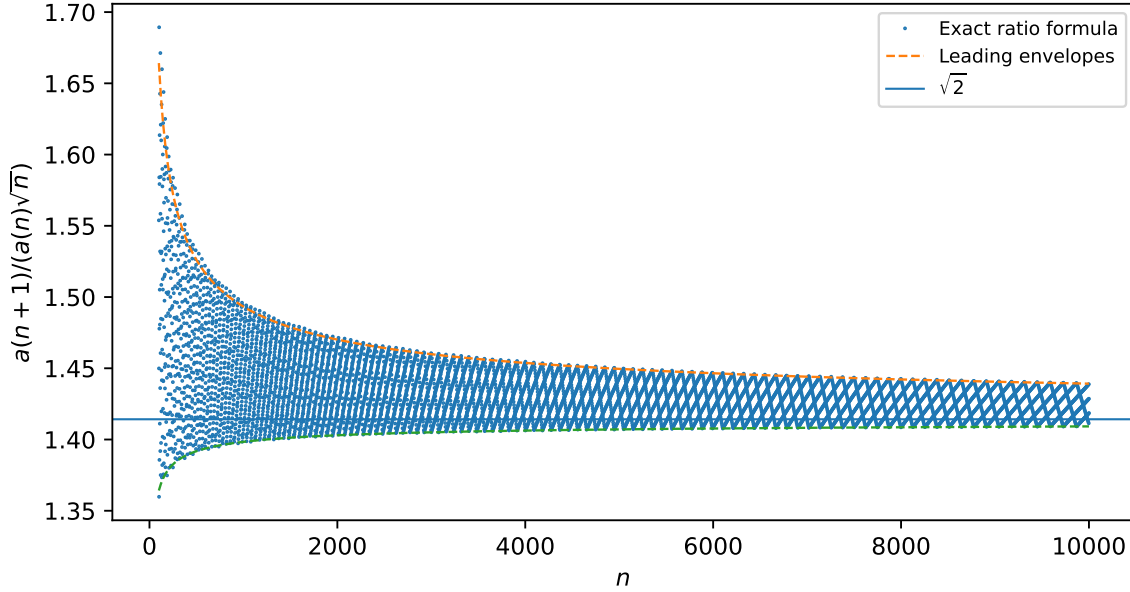


Figure 2: The ratio R_n for $100 \leq n \leq 10000$, computed from the exact transition formula. The horizontal line is $\sqrt{2}$. The other curves, $\sqrt{2} - 1/(2\sqrt{n})$ and $\sqrt{2} + 5/(2\sqrt{n})$, are leading asymptotic envelopes, not asserted finite- n inequalities.

8 Comparison with the maximum absolute coefficient

Let $b(n)$ denote A277281, the maximum coefficient magnitude [4]. Its candidate degrees satisfy only $d \equiv n \pmod{2}$. For consecutive candidates,

$$\frac{C_n(d+2)}{C_n(d)} = \frac{2(n-d)}{(d+1)(d+2)}. \quad (46)$$

The threshold is now the root of $d^2 + 5d + 2 - 2n = 0$, namely

$$\rho_n = \frac{\sqrt{8n+17} - 5}{2}.$$

Any maximizing exponent t_n lies in $[\rho_n, \rho_n + 2]$, with the obvious restriction to the finite allowed lattice. Ties can occur in this unsigned problem. For sufficiently large n ,

$$t_n = s_n - \frac{3}{2} + \varepsilon_n, \quad |\varepsilon_n| \leq 1 + O(s_n^{-1}).$$

The same Stirling argument therefore gives

$$b(n) \sim L(n), \quad \frac{a(n)}{b(n)} \longrightarrow 1. \quad (47)$$

More precisely, applying the uniform local expansion to both exponents,

$$\log \frac{a(n)}{b(n)} = \frac{\varepsilon_n^2 - \delta_n^2}{2s_n} + O(s_n^{-2}). \quad (48)$$

Thus the sign restriction has a real, rounding-dependent effect at relative order $n^{-1/2}$, but it does not change the leading constant. As a simple example of an unsigned tie, H_8 has coefficients 13440 and -13440 at degrees 4 and 2, respectively. The signed maximum still has the unique positive maximizing exponent from Theorem 2.1.

9 Algorithms, numerical checks, and reproducibility

9.1 A single exact term

The essential implementation is short:

```
from math import factorial, isqrt

def a277280(n: int) -> int:
    if isinstance(n, bool) or not isinstance(n, int):
        raise TypeError("n must be an integer")
    if n < 0:
        raise ValueError("n must be nonnegative")
    k = (2 * n + 6 - isqrt(8 * n + 21)) // 8
    d = n - 4 * k
    return (factorial(n) << d) // (factorial(2 * k) * factorial(d))
```

The integer square root is important: the decision is exact even when a floating-point approximation is too close to a floor boundary to be safe. The fuller implementation in `code/a277280.py` also accepts objects implementing Python's integer-index protocol and exposes the mode separately.

Finding the mode uses one integer square root and a bounded number of integer arithmetic operations on $O(\log n)$ -bit inputs. This is a count of operations, not a constant-time bit-complexity claim. Evaluating the coefficient still involves large-integer factorials, products, and a division. By (21), the output itself has $\Theta(n \log n)$ bits, so its construction and output cannot be constant-time in a bit model.

9.2 Streaming a prefix

Proposition 7.1 gives a prefix algorithm that does not recompute factorials and does not construct Hermite polynomials. Maintain (n, d, m, A) with $d = d_n$, $m = (n - d)/2$, and $A = a(n)$, initially $(0, 0, 0, 1)$. If $d(d + 1) < 2n + 7$, replace

$$A \leftarrow \frac{2(n+1)A}{d+1}, \quad d \leftarrow d+1,$$

leaving m unchanged. Otherwise replace

$$A \leftarrow \frac{(n+1)d(d-1)(d-2)A}{8(m+1)(m+2)}, \quad d \leftarrow d-3, \quad m \leftarrow m+2.$$

Then increment n . All divisions are exact. The implementation checks the remainder with `divmod`, so an implementation error cannot silently turn into a truncated integer value.

The algorithm uses $O(N)$ transitions for terms through N , each involving a bounded number of operations on the current large integer and factors of $O(\log N)$ bits. The total binary size of all

outputs is $\Theta(N^2 \log N)$, which is a more informative scale than a bare arithmetic operation count when writing a long b-file.

9.3 Numerical evaluation without enormous factorials

For large n , compute

$$\log a(n) = \log \Gamma(n+1) - \log \Gamma(m+1) - \log \Gamma(d+1) + d \log 2. \quad (49)$$

Subtract $\log L(n)$ at high precision, then apply `exp` or `expm1`. In particular, `expm1` avoids loss of relative accuracy when evaluating a correction close to zero. Guard precision is needed because the individual gamma logarithms have size $O(n \log n)$ while their residual after normalization can have size $O(n^{-1/2})$. The supplied routine adds guard digits based on the size of n .

Table 1 compares the leading approximation and its first two logarithmic corrections. The values were computed at high precision; the displayed decimals are rounded. The low index $n = 10$ is deliberately included to show that small- n accuracy need not improve monotonically with the addition of asymptotic terms.

n	d_n	$a(n)/L(n)$	$a(n)/L_1(n) - 1$	$a(n)/L_2(n) - 1$
10	2	0.9900563798	-2.490×10^{-1}	-9.401×10^{-2}
100	12	1.0959224302	-1.452×10^{-2}	-2.300×10^{-3}
1000	44	1.0320386856	1.151×10^{-4}	-8.533×10^{-6}
10000	140	1.0120994106	-3.108×10^{-5}	-9.717×10^{-7}
10^6	1,412	1.0010271211	-1.361×10^{-6}	-2.373×10^{-9}
10^8	14,140	1.0001065067	-1.221×10^{-8}	-2.194×10^{-12}
10^{12}	1,414,212	1.0000012066	-4.380×10^{-13}	-1.159×10^{-18}

Table 1: High-precision comparisons using the exact maximizing degree. L_j is defined by (33). More digits and the L_3 results appear in `data/asymptotic_comparison.csv`.

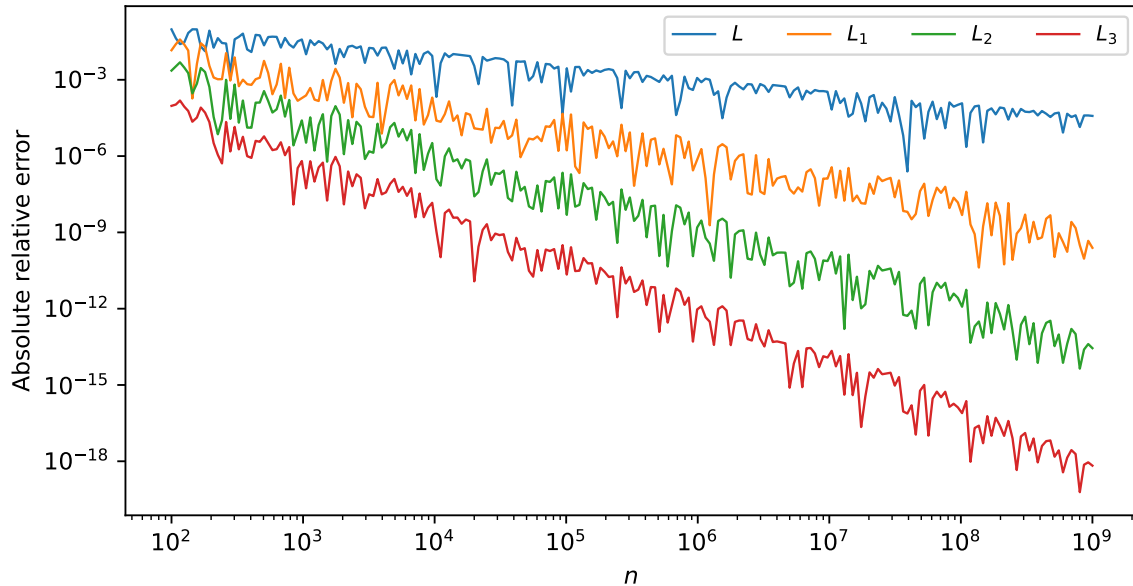


Figure 3: Absolute relative errors of L, L_1, L_2, L_3 at logarithmically spaced indices. Their proved uniform orders are, respectively, $n^{-1/2}, n^{-1}, n^{-3/2}, n^{-2}$. The varying correction polynomials cause the fluctuations and occasional near-cancellations.

9.4 What was actually checked

The included verification script completed the following independent checks. It matched the 25 terms $n = 0, \dots, 24$ displayed in the OEIS text entry; constructed H_n independently from $H_{n+1} = 2xH_n - 2nH_{n-1}$ for $n \leq 300$ and compared its maximum; and enumerated every positive coefficient for $n \leq 800$ using exact coefficient ratios. It also compared the streaming algorithm and the factorial formula for $n \leq 1000$.

Further integer checks verified the neighboring-mode inequalities for every $n \leq 100000$ and for 500 reproducibly sampled integers below 10^{100} , and verified the quartic factorization over all legal adjacent positive pairs for $n < 1000$. High-precision evaluations at eight selected indices through 10^{12} agreed with the analytic Stirling bounds. An additional SymPy calculation verified all three correction polynomials exactly as symbolic expressions.

These finite checks support the implementations and algebra, not the infinite asymptotic conclusions: those follow from the proofs above. The full external OEIS b-file was not included in the executed comparison; the script has an optional `-oeis-file` argument to check it. The bundled table through $n = 1000$ was generated independently by the streaming algorithm and is explicitly labeled as computed data, not as a downloaded OEIS reference file.

9.5 Archive contents and regeneration

The archive contains `article.tex` and its compiled PDF; the exact and high-precision library `code/a277280.py`; the independent test script `code/verify.py`; the table and figure generator `code/reproduce.py`; and the optional symbolic calculation `code/symbolic_check.py`. The `data` directory contains verification results, a computed b-file through 1000, and high-precision comparison

data. The `figures` directory contains the three plots in both PDF and PNG formats.

The exact algorithms use only the Python standard library. Reproducing the numerical figures requires `mpmath`, `numpy`, and `matplotlib`; the optional algebra check uses `sympy`. The README and Makefile give the complete regeneration commands. No network access is needed to rebuild the article or run the supplied checks.

10 Conclusion

The asymptotic growth of A277280 is fully determined by

$$a(n) \sim \frac{e^{-1/2}}{\sqrt{\pi} (2n)^{1/4}} \left(\frac{2n}{e}\right)^{n/2} e^{\sqrt{2n}}.$$

The key exact fact is that the maximizing positive degree is found by a single integer square root. Its location is $\sqrt{2n} - 3/2$ up to a bounded lattice-rounding displacement. Stirling's formula then determines not only the dominant growth, but also the leading constant and a complete scheme for further corrections.

The rounding displacement matters first at relative order $n^{-1/2}$. It explains why a smooth constant second term is unavailable, why the maximum absolute coefficient has the same leading asymptotic, and why the normalized consecutive ratios oscillate while converging to $\sqrt{2}$. All statements hold along the full sequence, without an unmentioned parity subsequence or averaging assumption.

References

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- [2] National Institute of Standards and Technology, *Digital Library of Mathematical Functions*, Section 18.5, equation 18.5.13: explicit finite expansion of H_n . <https://dlmf.nist.gov/18.5.E13>. Consulted 20 September 2026.
- [3] National Institute of Standards and Technology, *Digital Library of Mathematical Functions*, Section 5.11: asymptotic expansions of the gamma function; in particular equation 5.11.1 and Section 5.11(ii), remainder bounds. <https://dlmf.nist.gov/5.11>. Consulted 20 September 2026.
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